

ASSEMBLYLENS

Assembly Lens

Local First Mixed Reality Engineering Review on Meta Quest 3

DOCUMENT	Technical Case Study
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Abstract

Assembly Lens is a local-first mixed-reality engineering review system for Meta Quest 3. It imports CAD-derived GLB assemblies, preserves component identity and supports spatial review through selection, measurements, pins, visibility controls, exploded views, model manipulation, undo and persistent sessions. This case study describes the system architecture and a controlled investigation of an intermittent grey-view failure. Renderer and collider ablation separated visible geometry from collision-driven runtime behavior. Disabling 4,686 model colliders prevented the failure in repeated approaches; restoring them restored the failure. Source inspection then identified a model-root bounds trigger on a layer watched by Meta's wall-penetration comfort system. Moving the trigger to the ModelParts layer removed the false interaction in the reported fixed-build test.

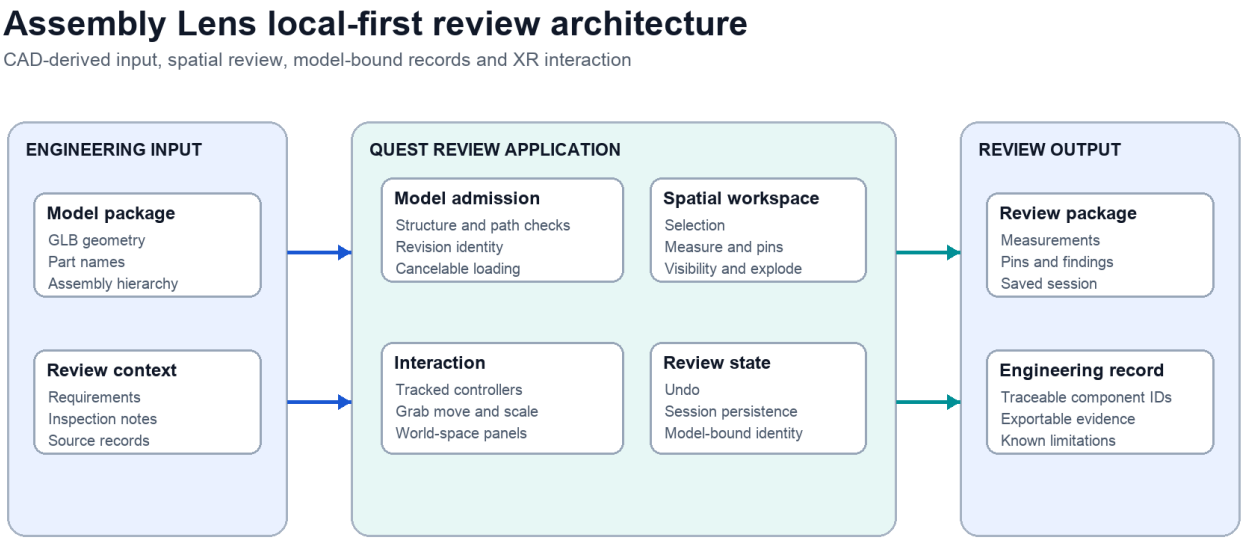
1 Engineering context

Industrial reviewers often need to inspect an assembly at life size, connect observations to individual components and preserve a record that another engineer can audit. Conventional model viewers frequently separate geometry, requirements and inspection evidence across different tools. Assembly Lens explores a spatial review workflow in which the model, the selected component and the review record remain synchronized inside a passthrough workspace.

2 System objectives

- Load a user-provided CAD-derived assembly without transferring the model to an Assembly Lens server.
- Preserve stable component identity across selection, hierarchy, measurements, pins and saved sessions.
- Support direct spatial inspection through tracked head and controller interaction.
- Retain review state locally and bind restoration to the exact model revision.
- Treat device behavior, Editor behavior and source-level checks as distinct forms of evidence.

3 Architecture



3.1 Model admission and identity

The model pipeline applies GLB structure and path checks before import, supports cancellation and assigns a revision identity used by persistence. Selection has one owner, which publishes changes to the hierarchy and contextual information. This prevents independent interface elements from maintaining conflicting selections.

3.2 Review interaction

The workspace combines model pose and scale control with component-level actions. Reviewers can isolate or reveal parts, create measurements, place numbered surface pins, adjust exploded views and reverse supported actions through a chronological undo history.

3.3 Local review records

Sessions preserve model pose, visibility, selection, pins and measurements in app-private storage. Restoration checks model revision before applying saved state. Trial launch uses Meta Platform entitlement; the engineering model and review data remain on-device.

4 Controlled grey view investigation

The room view intermittently became grey near a large assembly while the application continued to run and compositor-backed panels remained visible. The trigger occurred at a repeatable position relative to the model and moved when the model moved.

4.1 Competing hypotheses

The investigation considered frame-rate collapse, application pause, panel occlusion, camera clear behavior, submitted depth, passthrough loss, tracking behavior, visible model geometry and collision-driven comfort effects.

Controlled intervention	Observation	Inference
No model or initial small model	Grey did not reproduce	Model-associated spatial state was required
Move enlarged model	Trigger moved with the model	A fixed room location was excluded
Hide 4,685 renderers	Grey remained	Visible model rendering was not causal
Disable 4,686 colliders	No grey in three approaches and a separate confirmation	Collision state was necessary for the tested failure
Restore 4,686 colliders	Grey returned	Restoration strengthened the causal conclusion

4.2 Mechanism and correction

Imported part colliders were already on ModelParts layer 8. A large root BoxCollider remained on Default layer 0, which Meta XR Interaction's wall-penetration query watched through mask value 7. Enlarging the assembly enlarged this root box. A face of the box could therefore intersect the head-proximity ray in empty space and activate Meta's tunneling comfort effect beneath the application's compositor panels. The correction assigned only the root trigger to ModelParts while preserving model interaction and bounds behavior.

4.3 Evaluation result

Focused source and Editor checks passed, and the user reported no grey recurrence at the previous trigger locations in the fixed-APK trial. The result supports the identified mechanism for the reproduced failure. Sustained performance, thermals, stereo comfort and the full interaction checklist remain separate device acceptance tasks for each release candidate.

5 Engineering evaluation

Area	Result	Evidence scope
Safe regression baseline	28 suites and 538 passing assertions	Current source and Unity Editor
Grey-view causal test	Renderer and collider A-B-A investigation	Physical Quest 3 diagnostic build
Fixed-build observation	No recurrence at prior trigger locations	Physical Quest 3 user test
Historical diagnostic performance	Approximately 72 Hz during captured grey episodes	Pre-fix diagnostic build

6 Contributions

1. A local-first spatial review architecture that keeps component identity synchronized across the model, hierarchy and review record.
2. A reproducible experimental method for separating rendering, physics and vendor-runtime explanations in mixed reality.
3. A model-revision approach for preventing saved review state from being applied to the wrong engineering artifact.
4. A practical research platform for studying evidence-grounded multimodal assistance and calibrated human deferral.

7 Limitations and next work

- The current Alpha candidate requires a complete physical Quest pass for controller behavior, sustained 72 Hz operation, thermals, memory and stereo clarity.
- Decoded texture memory, Draco working memory and final GPU allocation are not fully bounded by JSON preflight.
- The current application does not implement the multimodal assistant described in the accompanying research proposal.
- The local Trial record is suitable for invited pilots, not tamper-resistant commercial licensing.

References

[1] Assembly Lens Technical Evaluation Report, 17 October 2026. [2] Assembly Lens Grey-Screen Root-Cause Summary, 4 October 2026. [3] Unity Technologies, Physics.Raycast documentation, Unity 2022.3. [4] Khronos Group, glTF 2.0 Specification.